

Problem

Given that $v(0) = 5$ and $dv(0)/dt = 10$, solve

$$\frac{d^2v}{dt^2} + 5\frac{dv}{dt} + 6v = 25e^{-t}u(t)$$

Step-by-step solution

Step 1 of 5

The differential equation is,

$$\frac{d^2v}{dt^2} + 5\frac{dv}{dt} + 6v = 25e^{-t}u(t)$$

Apply Laplace transform.

$$\left[s^2V(s) - sv(0) - v'(0)\right] + 5\left[sV(s) - v(0)\right] + 6V(s) = \frac{25}{s+1}$$

Substitute 5 for $v(0)$ and 10 for $\frac{dv(0)}{dt}$ in the above equation.

$$s^2V(s) - s(5) - 10 + 5[sV(s) - 5] + 6V(s) = \frac{25}{s+1}$$

$$(s^2 + 5s + 2)V(s) - 5s - 10 - 25 = \frac{25}{s+1}$$

$$(s^2 + 5s + 2)V(s) = \frac{25}{s+1} + 5s + 35$$

$$(s^2 + 5s + 2)V(s) = \frac{25 + 5(s+7)(s+1)}{s+1}$$

Step 2 of 5

On further simplification:

$$V(s) = \frac{25 + 5s^2 + 40s + 35}{(s+1)(s^2 + 5s + 2)}$$

$$= \frac{5s^2 + 40s + 60}{(s+1)(s^2 + 5s + 2)}$$

$$= \frac{5s^2 + 40s + 60}{(s+1)(s+2)(s+3)}$$

Apply partial fractions to simplify the above equation.

$$V(s) = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3} \dots\dots (1)$$

Step 3 of 5

Calculate the constant A using residue theorem.

$$\begin{aligned} A &= (s+1)V(s)\Big|_{s=-1} \\ &= (s+1)\frac{5s^2+40s+60}{(s+1)(s+2)(s+3)}\Big|_{s=-1} \\ &= \frac{5(-1)^2+40(-1)+60}{(-1+2)(-1+3)} \\ &= 12.5 \end{aligned}$$

Calculate the constant B using residue theorem.

$$\begin{aligned} B &= (s+2)V(s)\Big|_{s=-2} \\ &= (s+2)\frac{5s^2+40s+60}{(s+1)(s+2)(s+3)}\Big|_{s=-2} \\ &= \frac{5(-2)^2+40(-2)+60}{(-2+1)(-2+3)} \\ &= \frac{20-80+60}{(-1)(1)} \\ &= 0 \end{aligned}$$

Step 4 of 5

Calculate the constant C using residue theorem.

$$\begin{aligned} C &= (s+3)V(s)\Big|_{s=-3} \\ &= (s+3)\frac{5s^2+40s+60}{(s+1)(s+2)(s+3)}\Big|_{s=-3} \\ &= \frac{5(-3)^2+40(-3)+60}{(-3+1)(-3+2)} \\ &= \frac{45-120+60}{(-2)(-1)} \\ &= -7.5 \end{aligned}$$

Step 5 of 5

Recall equation (1).

$$V(s) = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3}$$

Substitute 12.5 for A , 0 for B and -7.5 for C in the above equation.

$$\begin{aligned} V(s) &= \frac{12.5}{s+1} + \frac{0}{s+2} + \frac{-7.5}{s+3} \\ &= \frac{12.5}{s+1} - \frac{7.5}{s+3} \end{aligned}$$

It is known that,

$$L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$$

Apply inverse Laplace transform.

$$\begin{aligned} v(t) &= 12.5e^{-t}u(t) - 7.5e^{-3t}u(t) \\ &= [12.5e^{-t} - 7.5e^{-3t}]u(t) \end{aligned}$$

Therefore, output, $v(t)$ of the differential equation is $\boxed{[12.5e^{-t} - 7.5e^{-3t}]u(t)}$.